

5- Theory - HW

1 Encrypt

PT = Enjoy your Journey d = 2, 3

d = 2

E	J	Y	O	R	O	R	E
N	O	Y	U	J	U	N	Y

CT = E J Y O R O R E N O Y U J U N Y

d = 3

E		Y		R		R		X
N		O		Y		U		J
	J		O		O		E	

CT = E Y R R X N O Y U J U N Y J O O E

2 Message :- Print Screen

d = 2

P	I	T	C	E	N
R	N	S	R	E	

CT = P I T C E N

R N S R E

d = 3

P		T		E	
R	N	S	R	E	
	I		C		N

CT = P T E R N S R E - I S N

3 Decrypt :- E O C S N Y U N L E J R E A S O N S d = 4

L = 19

19/4 = 4

E				O					C				S
N			Y		U			N		L		E	
	J		Y		R		E			A	S		
		O				N				S			

PT = ENJOY YOUR NEW CLASSES

4 Decrypt BUOI LUYUBOSNVU TYRKAS $d=3$

B		U		O						I			
	V		Y		V	B		O	S	N		V	
		Y			R			K		A			

PT = BUY YOUR BOOKS IN AUGUST

V		T
	S	

5 SEMSERTISINCMOO $d=2$

S E M S E R T I
S I N C M O O

PT = SSEIMN SCEMROTOI

Permutation :-

1 Message = WE are discovered Key = 53124

5 3 1 2 4
W E A R E
D I S L O
V E R E D

CT = ~~W~~ A S R R C E E I E E O D W O V

DFS

1 ZIP in hexadecimal : 0x0002000000000001

Initial permutation in given table

0000	0000	0000	0010	bit position
0010	0000	0000	0000	
0000	0000	0000	0000	64 → 25
0000	0000	0000	0001	

Apply initial permutation

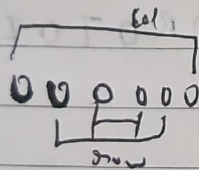
0000	0000	0000	0000	0000	0000	1000	0000
0000	0000	0000	0000	0000	0000	0000	0010

Hexa: 0x00 0004 0000 0001

2. $M = 1101$ $K = 1011$ $\rightarrow$ XOR

$$\begin{array}{r} 1101 \\ + 1011 \\ \hline 0110 \end{array} \quad CT = 0110$$

3. ZIP $\rightarrow$ 5 box 1 = 000000



$C = 0$, $R = 0$ in 5 box 1 = 14

$O/P = 01110$

6-bit register

270